Lesson One

Applying to Stellinga College

Why Stellinga?
Thank you for your interest in Stellinga International College. As an international student, we are sure you will find our university college an exciting place to study, with like-minded and ambitious individuals.

Preparing and submitting your application
We have tried to make the application process as easy as possible for you, but there are a number of procedures you must follow.

All our courses are taught in English, so first of all you will probably need to submit proof of your English language ability. We require an IELTS score of 6.5 or another test result which is equivalent. You will also have to send us your secondary school diploma, so that we could can evaluate it. If you have the International Baccalaureate or a Dutch VWO diploma, you don't need to provide English language test results.

We will also require a personal statement. This is a text of up to 1,000 words in which you introduce yourself, explain your interest in our college, and why you want to study your chosen course.

If you are from outside the European Union (EU), it is important that you have an entrance visa before you come to study in the Netherlands, but we will apply for this for you.

We now play accept online applications, so please ensure that you have all your documents ready to upload before you begin. Any documents that are not in English originally will also need to be translated and the translations also uploaded. You will need a passport photograph and passport copy.

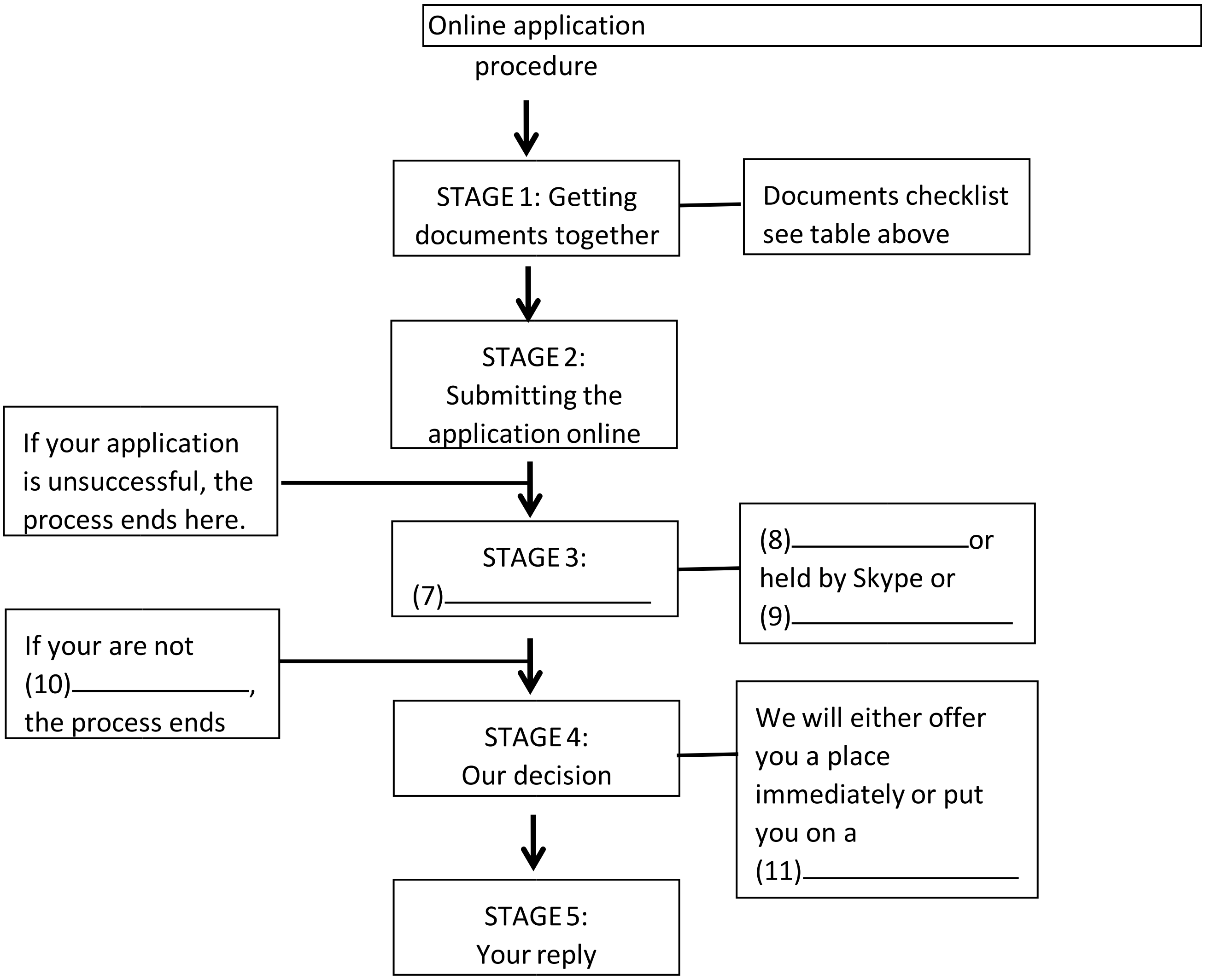
What happens next?
Your application will then be considered. If your initial application is successful, you will be invited for an interview. This will be conducted in English via Skype, over the phone or on site, depending on whether you can come and visit us. You will talk to two or three members of staff for up to 30 minutes, and will be asked to elaborate on your application documents and your personal statement. We aim to inform you of our decision in writing, within 4 weeks. There are several possible outcomes: you may not have been successful; you may be offered a place at the college or you may be offered a place on the waiting list. You will need to reply to any offers within two weeks, otherwise your place may be offered to somebody else.
Good luck with your application.

Questions 1-11

Using *NO MORE THAN TWO WORDS* from the passage for each answer, complete the table and the flow chart below. The required documents:

Evidence of language ability	IELTS 6.5 or (1)
Evidence of studies	(2)____________, Dutch VWO diploma, or other secondary school diploma
Information about motivation	(3) ____________with a maximum length of (4)
Proof of identity	(5) ____________and passport photo
Other	(6) ____________if originals are in a foreign language

The online application process for people outside the EU:



Lesson Two

长难句分析 1) 划分句子 2) 划出主干、并列、从句

1. Automated techniques to measure these characteristics and verify a person’s identify infallibly are attracting widespread attention in these days of increased use of credit card and e-mail.

2. His scheme would employ up to 16 trillion minute spacecraft, each weighing about one gram, to form a transparent, sunlight-refracting sunshade in an orbit 1.5 million km above the Earth.

3. The problem of how health-care resources should be allocated or apportioned, so that they are distributed in both the most just and most efficient way, is not a new one.

4. In the Native American Navajo nation, which sprawls across four states in the American south-west, the native language is dying.

5. Such developments have been criticised as an intolerable vulgarisation, but the success of many historical theme parks and similar locations suggests that the majority of the public does not share this opinion.

READINGPASSAGE1

You should spend about 20 minutes on ***Question 1-13*** *which are based on Reading Passage 3 below.*

Stepwells

A millennium ago, stepwells were fundamental to life in the driest oarts of India. Richard Cox travelled to north-western India to document these spectacular monuments from a bygone era

During the sixth and seventh centuries, the inhabitants of the modern-day states of Gujarat and Rajasthan in north-western India developed a method of gaining access to clean. Fresh groundwater during the dry season for drinking, bathing, watering animals and irrigation. However, the significance of this invention - the stepwell goes beyond its utilitarian application.

Unique to this region, stepwells are often architecturally complex and vary widely in size and shape. During their heyday, they were places of gathering, of leisure and relaxation and of worship for villagers of all but tho lowest classes. Most stepwells are found dotted round the desert areas of Gujarat (where they are called *vav*) and Rajasthan (where they are called *baori*), while a few also survive in Delhi. Some wore located in or near villages as public spaces for the community; others were positioned beside roads as resting places for travellers.

As their name suggests, stepwells comprise a series of stone steps descending from ground level to the water source (normally an underground aquifer) as it recedes following the rains. When the water level was high, the user needed only to descend a few steps to reach it; when it was low, several levels would have to be negotiated.

Some wells are vast, open craters with hundreds of steps paving each sloping side, often in tiers. Others are more elaborate, with long stepped passages leading to the water via several storeys. Built from stone and supported by pillars, they also included pavilions that sheltered visitors from the relentless heat. But perhaps the most impressive features are the intricate decorative sculptures that embellish many stepwells, showing activities from fighting and dancing to everyday acts such as women combing their hair or churning butter.

Down the centuries, thousands of wells were constructed throughout north-western India, but the majority have now fallen into disuse; many are derelict and dry, as groundwater has been diverted for industrial use and the wells no longer reach the water table. Their condition hasn’t been helped be recent dry spells: southern Rajasthan suffered an eight-year drought between 1996 and 2004.

However, some important sites in Gujarat have recently undergone major restoration, and the state government announced in June last to restore the stepwells state.

In Patan, the state's ancient capital, the stepwell of *Rani Ki Vau* (Queen's Stepwell) is perhaps the finest current example. It was built by Queen Udayamati during the late 11th century, but became silted up following a flood during the 13th century. But the Archaeological Survey of India began restoring it in the 1960s, and today it is in pristine condition. At 65 meters long, 20 meters wide and 27metres deep, Rani Ki Vau features 500 sculptures carved into niches throughout the monument. Incredibly, in January 2001, this ancient structure survived an earthquake that measured 7. 6 on the Richter scale.

Another example is the *Surya Kund* in Modhera, northern Gujarat, next to the Sun Temple, built by King Bhima I in 1026 to honour the sun god Surya. It actually resembles a tank (*kund* means reservoir or pond) rather than a well, but displays the hallmarks of stepwell architecture, including four sides of steps that descend to the bottom in a stunning geometrical formation. The terraces house 108 small, intricately carved shrines between the sets of steps.

Rajasthan also has a wealth of wells. The ancient city of Bundi, 200 kilometers south of Jaipur, is renowned for its architecture, including its stepwells. One of the larger examples is *Raniji Ki Baori*, which was built by the queen of the region, Nathavatji, in 1699. At 46 meters deep, 20 meters wide and 40 meters long, the intricately carved monument is one of 21 *baoris* commissioned in the Bundi area by Nathavatji.

In the old ruined town of Abhaneri, about

95 kilometres east of Jaipur, is *Chand Baori*, one of India's oldest and deepest wells; aesthetically it's perhaps one of the most dramatic. Built in around 850 AD next to the temple of Harshat Mata, the baori comprises hundreds of zigzagging steps that run along three of its sides, steeply descending 11 storeys, resulting in a striking pattern when seen from afar. On the fourth side, verandas which are supported by ornate pillars overlook the steps.

Still in public use located just off the Jaipur-Delhi highway. Constructed in around 1700, it is nine storeys deep, with the last two being underwater. At ground level, there are 86 colonnaded openings from where the visitor descends 170 steps to the deepest water source.

Today, following years of neglect, many of these monuments to medieval engineering have been saved by the Archaeological Survey of India, which has recognized the importance of preserving them as part of the country's rich history. Tourists flock to wells in far-flung corners of north-western India to gaze in wonder at these architectural marvels from hundreds of years ago, which serve as a reminder of both the ingenuity and artistry of ancient civilizations and of the value of water to human existence.

Question 9-13

Complete the table below.

Choose ***ONE WORD AND/OR A NUMBER*** *from the passage for each answer.*

Write your answers in boxes 9-13 on your answer sheet.

Stepwell	**Date**	**Features**	**Other notes**
Ran Ki Vav	Late 11th century	As many as 500 sculptures decorate the monument	Restored in the 1960s Excellent condition, despite the **9** __________ of 2001
Surya Kund	1026	Step on the **10** __________ produce a geometrical pattern Carved shrines	Looks more like a **11** __________ than a well
Raniji Ki Baori	1699	Intricately carved monument	One of 21 *baoris* in the area commissioned by Queen Nathavatji
Chand Baori	850AD	Steps take you down 11 storeys to the bottom	Old, deep and very dramatic Has **12** __________ which provide a view of the steps
Neemrana Ki Baori	1700	Has two **13** __________ levels	Used by public today

Section 2 Question 15-27

Read the text below and answer Question 15-20

Professional Credentials:
Advice for Immigrants

As an immigrant to North America, you will need to ensure that employers and organisations such as colleges and universities properly recognise your international credentials. These may be trade certificates, but also educational qualifications such as degrees.or diplomas, that you have completed or partially-completed.

It is common for hiring personnel to have little or no training in evaluating an academic background earned outside of North America. But at the same time, employers see formal education as very important when hiring. Education is a hiring requirement for 60% of employment opportunities/ but 40% of human resources staff say that if they do not know a lot about the value of documents attained elsewhere, they will not recognise them.

Research has shown that sometimes immigrants start with a lower salary level than people who have completed their training in North America. You may want to apply for employment opportunities with companies whose staff understands your situation or, more importantly, who know where to send you to get your North American qualifications. If you need to complete your training in North America, apprenticeships leading to skilled trades are in high demand. Apprenticeship training is a hands-on program where about 10% is in a classroom setting at community colleges, and 90% of the training is on-the-job. The training involves working for an employer and earning income during the training period. Sometimes there is a limit of 5 years for training. You may be able to use this training toward college or university credits or education. There is a good potential for long-term job security after completion of apprenticeship training.

If you earned your papers outside of North America, you will need to get them translated if you want to work or study. It is important for you that your education is assessed by an accredited assessment service when you are applying for jobs, and particularly if the job posting has an education requirement. As well, it is recommended that you include a copy of the report with your cover letter. It is suggested that you provide this information early and do not wait until the time you actually meet with the employer. Getting job interviews is more than 50% of the whole process of securing employment; and with an evaluation report ,you want to make sure that employers are screening you ‘in’ rather than ‘out’.

Establishing yourself North America is a difficult process but companies do consider integrating immigrants into the workforce important to the workplace mosaic. Employers are making significant progress in improving diversity at work.

Question 15-20

Complete the sentences below.

Choose ***NO MORE THAN TWO WORDS*** *from the text for each answer.*

Write your answers in boxes 15-20 on your answer sheet.

15 New arrivals to North America need to make sure that their academic qualifications or their ____________ are accepted.

16 A significant number of companies view ____________ as a major requirement.

17 People educated in North America may initially be offered a higher ____________ than immigrants.

____________ **18** courses often provide more job stability.

19 Most of the effort to find work is spend trying to obtain ____________.

20 As more newcomers enter the workforce, ____________ increases.

Read the text below and answer ***Questions 21-27***

How to prepare for a presentation

The first time your boss suggests that you formally present something to you department or a client, your reaction may be to panic. But remember that being asked to present is a compliment. Someone believes that you have valuable information to share with the group, and wants to listen to your ideas.

You need to decide exactly what you will say during the allotted time. Condense your topic into one sentence. What do you want your audience to remember or learn from your talk? This is your 'big idea'. Remember that you are dealing with the short attention spans of individual who tend to have many things on their mind.

Think of three main points you want to make to support your overall topic. Develop a story to demonstrate each of those concepts. This could be something that happened to you or someone you know, or something you read in a newspaper or magazine.

We have all heard the saying *A Picture is worth a thousand words*. Think about how your presentation can be more interesting to watch. Props are a wonderful way to make your talk come alive. You could do something as simple as holding up a toy phone receiver when talking about customer service or putting on a hat to signal a different part of your talk.

Think of a dynamic and unusual way to start your presentation. This might involve telling anecdotes that relate to your topic. Never begin with, ‘Thank you for inviting me here to talk with you today.’ You will put your audience to sleep right away. Start off enthusiastically so they will listen with curiosity and interest. After your energetic introduction, identify yourself briefly and thank the audience for taking the time to listen to you.

Plan your ending, and finish in a memorable way. Your listeners remember best what they heat at the beginning and end of a speech, so conclude with a game in which they can participate, or tell a humorous story and your audience will leave laughing.

Don’t try to memorise your talk or read it word-for-word. It will sound stilted and boring. Instead, practice your dynamic introduction and conclusion until you can deliver them effortlessly. If you do this you’ll feel a burst of confidence that will help you sail through the whole of the speech.

Question 21-27

Complete the sentences below.

*Choose **ONE WORD ONLY** from the text for each answer.*

Write your answers in boxes 21-27 on your answer sheet.

How to Prepare for a Presentation

- You should regard an invitation to speak as a 21 ____ .
- Express your main idea in a 22____________ .
- Try using a 23___ to support the major points you are making.
- Add visual excitement to your talk by using 24 . ______________
- Express appreciation to your listeners for their 25. __________________
- A 26________________ will get the audience to interact.
- It is important to prepare well as this will increase your 27 ___________________.

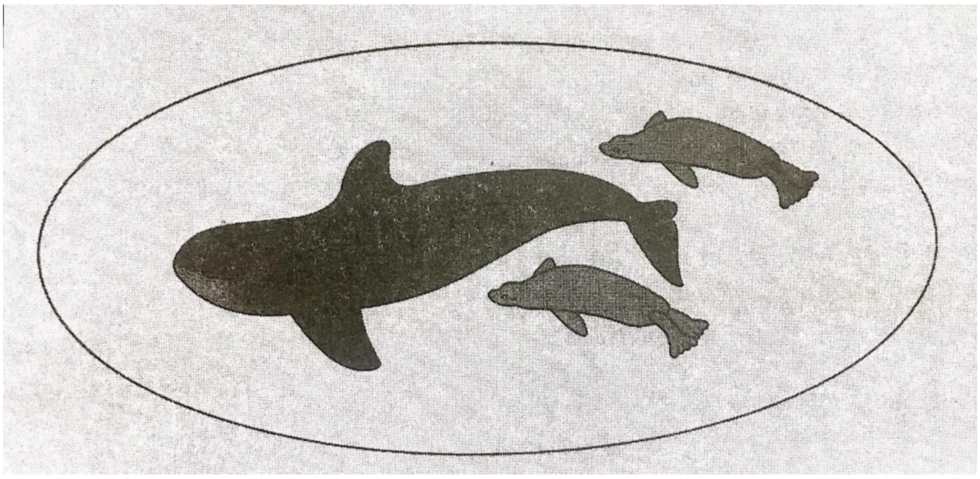
Lesson Four

READING PASSAGE 2

You should spend about 20 minutes on ***Question 15-26*** *which are based on Reading Passage 2 below.*

What Do Whales Feel?

An examination of the functioning of the senses in cetaceans, the group of mammals comprising whales, dolphins and porpoises.



Some of the senses that we and other terrestrial mammals take for granted are either reduced or absent In cetaceans or fail to function well in water. For example, It appears from their brain structure that toothed species are unable to smell. Baleen species, on the other hand, appear to have some related brain structures but it is not known whether these are functional. It has been speculated that, as the blowholes evolved and migrated to the top of the head, the neural pathways serving sense of smell may have been nearly all sacrificed. Similarly, although at least some cetaceans have taste buds, the nerves serving these have degenerated or are rudimentary.

The sense of touch has sometimes been described as weak too, but this view is probably mistaken. Trainers of captive dolphins and small whales often remark on their animals' responsiveness to being touched or rubbed, and both captive and free ranging cetacean individuals of all species (particularly adults and calves, or members of the same subgroup) appear to make frequent contact. This contact may help to maintain order within a group, and stroking or touching are part of the courtship ritual in most species. The area around the blowhole is also particularly sensitive and captive animals often object strongly to being touched there.

The sense of vision is developed to different degrees in different species. Baleen species studied at close quarters underwater-specifically a grey whale calf in captivity for a year, and free-ranging right whales and humpback whales studied and filmed off Argentina and Hawaii-have obviously tracked objects with vision underwater, and they can apparently see moderately well both in water and in air. However, the position of the eyes so restricts the field of vision in baleen whales that they probably do not halve stereoscopic vision.

On the other hand, the position of the eyes In most dolphins and porpoises suggests that they have stereoscopic vision forward and downward. Eye position in freshwater dolphins, which often swim on their side or upside down while feeding, suggests that what vision they have ls stereoscopic forward and upward. By comparison, the bottlenose dolphin has extremely keen vision In water. Judging from the way it watches and tracks airborne flying fish, it can apparently see fairly well through the air- water interface as well. And although preliminary experimental evidence suggests that their in-air vision is poor, the accuracy with which dolphins leap high to take small fish out of a trainer's hand provides anecdotal evidence to the contrary.

Such variation can no doubt be explained with reference to the habitats in which individual species have developed. For example, vision is obviously more useful to species inhabiting clear open waters than to those living in turbid rivers and flooded plains. The South American boutu and Chinese beiji, for instance, appear to have very limited vision, and the Indian susus are blind, their eyes reduced to slits that probably allow them to sense only the direction and intensity of light.

Although the senses of taste and smell appear to have deteriorated, and vision in water appears to be uncertain, such weaknesses are more than compensated for by cetaceans' well-developed acoustic sense. Most species are highly vocal, although they vary in the range of sounds they produce, and many forage for food using echolocation. Large baleen whales primarily use the lower frequencies and are often limited in their repertoire. Notable exceptions are the nearly song-like choruses of bowhead whales in summer and the complex, haunting utterances of the humpback whales. Toothed species in general employ more of the frequency spectrum, and produce a wider variety of sounds, than baleen species (though the sperm whale apparently produces a monotonous series of high-energy clicks and little else). Some of the more complicated sounds are clearly communicative, although what role they may play in the social life and 'culture' of cetaceans has been more the subject of wild speculation than of solid science.

Questions 15-21

Complete the table below.

Choose **NO MORE THAN THREE WORDS** *from Reading Passage 2 for each answer.*

Write your answers in boxes 15-21 on your answer sheet.

SENSE	SPECIES	ABILITY	COMMENTS
Smell	toothed	no	Evidence from brain structure
	baleen	not certain	Related brain structure are present
Taste	Some type	poor	Nerves linked to their **15** ______ are underdeveloped
Touch	all	yes	Region around the blowhole very sensitive
Vision	**16** __________	yes	Probably do not have stereoscopic vision
	Dolphins, porpoises	yes	Probably have stereoscopic vision **17** ______ and ______
	18 __________	yes	Probably have stereoscopic vision forward and upward
	bottlenose dolphins	yes	Exceptional in **19** __________ and good in air-water interface
	Boutu and beiji	poor	Have limited vision
	Indian susus	no	Probably only sense direction and intensity of light
Hearing	Most large baleen	yes	Usually use **20** _________; Repertoire limited
	21 whales ______ and ___________ whales	yes	Song-like
	toothed	yes	Use more of frequency spectrum; have wider repertoire

Questions 22-26

Complete the table below.

Answer the questions below using ***NO MORE THAN THREE WORDS*** *from the passage for each answer.*

Write your answers in boxes 22-26 on your answer sheet.

22 Which of the sense is described here as being involved in mating?

23 Which species swims upside down while eating?

24 What can bottlenose dolphins follow from under the water?

25 Which type of habitat is related to good visual ability?

26 Which of the senses is best developed in cetaceans?

<u>**Lesson Five**</u>

READING PASSAGE 2

*You should spend about 20 minutes on **Question 14-27** which are based on Reading Passage 2 on the following pages.*

Question 14-18

Reading Passage 2 contains six Key Points.

Choose the correct heading for Key Points **TWO** to **SIX** from the list of heading.

Write the correct number, i-viii, in boxes 14-18 on your answer sheet.

List of Headings

i	Ensure the reward system is fair
ii	Match rewards to individuals
iii	Ensure targets are realistic
iv	Link rewards to achievement
v	Encourage managers to take more responsibility
vi	Recognise changes in employees' performance over time
vii	Establish targets and give feedback
viii	Ensure employees are suited to their jobs

Example	*Answer*
Key Point **One**	**viii**

14 Key Point **Two**

15 Key Point **Three**

16 Key Point **Four**

17 Key Point **Five**

18 Key Point **Six**

Motivating Employees under
Adverse Conditions

THE CHALLENGE

It is a great deal easier to motivate employees in a growing organisation than a declining one. When organisations are expanding and adding personnel, promotional opportunities, pay rises, and the excitement of being associated with a dynamic organisation create feelings of optimism. Management is able to use the growth to entice and encourage employees.When an organisation is shrinking, the best and most mobile workers are prone to leave voluntarily.Unfortunately,they are the ones the organisation can least afford to lose-those with the highest skills and experience.The minor employees remain because their job options are limited.

KEY POINT ONE

There is an abundance of evidence to support the motivational benefits that result from carefully matching people to jobs. For example, if the job is running a small business or an autonomous unit within a larger business, high achievers should be sought. However, it the job to be filled is a managerial post in a large bureaucratic organisation, a candidate who has a high need for power and a low need for affiliation should be selected. Accordingly, high achievers should not be put into jobs that are inconsistent with their needs. High achievers will do best when the job provides moderately challenging goals and where there is independence and feedback. However, it should be remembered that not everybody is motivated by jobs that are high in independence,variety and responsibility.

KEY POINT TWO

The literature on goal-setting theory suggests that managers should ensure that all employees have specific goals and receive comments on how well they are doing in those goals. For those with high achievement needs, typically a minority in any organisation, the existence of external goals is less important because high achievers are already internally motivated. The next factor to be determined is whether the goals should be assigned by a manager or collectively set in conjunction with the employees. The answer to that depends on perceptions of goal acceptance and the organisation's culture. If resistance to goals is expected, the use of participation in goal-setting should increase acceptance.If participation is inconsistent with the culture, however, goals should be assigned. If participation and the culture are incongruous, employees are likely to perceive the participation process as manipulative and be negatively affected by it.

KEY POINT THREE

Regardless of whether goals are achievable or well within management's perceptions of the employee's ability, if employees see them as unachievable they will reduce their effort. Managers must be sure, therefore, that employees feel confident that their efforts can lead to performance goals. For managers, this means that employees must have the capability of doing the job and must regard the appraisal process as valid.

KEY POINT FOUR

Since employees have different needs, what acts as a reinforcement for one may not for another. Managers could use their knowledge of each employee to personalise the rewards over which they have control. Some of the more obvious rewards that managers allocate include pay, promotions, autonomy, job scope and depth, and the opportunity to participate in goal-setting and decision-making.

KEY POINT FIVE

Managers need to make rewards contingent on performance. To reward factors other than performance will only reinforce those other factors. Key rewards such as pay increases and promotions or advancements should be allocated for the attainment of the employee's specific goals. Consistent with maximising the impact of rewards, managers should look for ways to increase their visibility. Eliminating the secrecy surrounding pay by openly communicating everyone;s remuneration, publicising performance bonuses and allocating annual salary increases in a lump sum rather than spreading them out over an entire year are examples of actions that will make rewards more visible and potentially more motivating.

KEY POINT SIX

The way rewards are distributed should be transparent so that employees perceive that rewards or outcomes are equitable and equal to the inputs given. On a simplistic level, experience, abilities, effort and other obvious inputs should explain differences in pay, responsibility and other obvious outcomes. The problem, however, is complicated by the existence of dozens of inputs and outcomes and by the fact that employee groups place different degrees of importance on them. For instance, a study comparing clerical and production workers identified nearly twenty inputs and outcomes. The clerical workers considered factors such as quality of work performed and job knowledge near the top of their list, but these were at he bottom of the production workers' list. Similarly, production workers thought that the most important inputs were intelligence and personal involvement with task accomplishment, two factors that were quite low in the importance ratings of the clerks. There were also important, though less dramatic, differences on the outcome side. For example, production workers rated advancement very highly, whereas clerical workers rated advancement in the lower third of their list. Such findings. suggest that one person's equity is another's inequity, so an ideal should probably weigh different inputs and outcomes according to employee group.

The world's friendliest city

A team of social psychologists from California has spent six years studying the reactions of people in cities around the world to different situations. The results show that cities where people have less money generally have friendlier populations. Rio de Janeiro in Brazil, which is often known for its crime, comes out top, and the capital of Malawi, Lilongwe, comes third.

But what makes one city friendlier than another? The psychologists from California State University say it has got more to do with environment than culture or nationality.

They carried out a study into the way locals treated strangers in 23 cities around the world. The team conducted their research through a series of tests, where they dropped pens or pretended they were blind and needed help crossing the street.

The study concludes that people are more helpful in cities with a more relaxed way of life such as Rio. While they were there, researchers received help in 93 percent of cases, and the percentage in Lilongwe was only a little lower, However, richer cities such as Amsterdam and New York are considered the least friendly.

Inhabitants of Amsterdam helped the researchers in 53 percent of cases and in New York just 44 percent. The psychologists found that, in these cities, people tend to be short of time, so they hurry and often ignore strangers.

Table completion

*Complete the table. Choose **ONE WORD ONLY** from the passage for each answer.*

city	positive aspects	Negative aspects	% of help received
Rio de Janeiro	● friendly inhabitants ● more ______ lifestyle	● People don't have so much ____________ ● Has reputation for ____________	93%
Amsterdam and New York	● richer	● People… ● have little ; ● don't pay attention to ________________;	Amsterdam:53% New York:44%

The life of the European bee-eater

A brilliant movement of colour as it catches its food in the air, the European bee-eater moves between three continents.

True to their name, bee-eaters eat bees (though their diet includes just about any flying insect.) When the bird catches a bee, it returns to its tree to get rid of the bee's poison, which it does very efficiently. It hits the insect's head on one side of the branch, then rubs its body on the other. The rubbing makes its prey harmless.

European bee-eaters (Merops apiaster) form families that breed in the spring and summer across an area that extends from Spain to Kazakhstan. Farmland and river valleys provide huge numbers of insects. Flocks of bee-eaters follow tractors as they work fields. When the birds come upon a beehive, they eat well-a researcher once found a hundred bees in the stomach of a bee-eater near a hive.

European bees pass the winter by sleeping in their hives, which cuts off the bee-eater's main source of food. So, in late summer, bee-eaters begin a long, dangerous journey. Massive flocks from Spain, France and northern Italy cross the Sahara desert to their wintering grounds in West Africa. Bee-eaters from Hungary and other parts of Central and Eastern Europe cross the Mediterranean Sea and Arabian Desert to winter in southern Africa. 'It's an extremely risky stratagem, this migration,' says C. Hilary Fry, a British ornithologist who has studied European bee-eaters for more than 45 years. 'At least 30 percent of the birds will be killed by predators before they make it back to Europe the following spring.'

In April, they return to Europe. Birds build nests by digging tunnels in riverbanks. They work for up to 20 days. By the end of the job, they've moved 15 to 26 pounds of soil-more than 80 times their weight.

The nesting season is a time when families help each other, and sons or uncles help feed their father's or brother's chicks as soon as they come out of their eggs. The helpers benefit, too: parents with helpers can provide more food for chicks to continue the family line.

It's a short, spectacular life. European bee-eaters live for five to six years. The difficulties of migration and avoiding predators along the way affect every bird. Bee-eaters today also find it harder to find food, as there are fewer insects around as a result of pesticides. Breeding sites are also disappearing, as rivers are turned into concrete-walled canals.

Sentence completion

Complete sentences. Write ***NO MORE THAN TWO WORDS*** *from the passage for each answer.*

1. Bee-eaters' prey are bees and other ___.______
2. Bee-eaters need to remove the ____________ from bees before eating them.
3. There is plenty of food for bee-eaters on agricultural land and in_____.__________
4. Bee-eaters migrate to spend the winter in different parts of .
5. Because of _____________, almost one-third of bee-eaters do not survive migration.
6. Bee-eaters make nests in__________,which they build themselves.
7. When nesting, the______________receive food from different family members.

One problem for bee-eaters is , __________which have reduced the amount of food available.

Mau Piailug, ocean navigator

Mau sailed from Hawaii to Tahiti using traditional methods. In early 1976, Mau Piailug, a fisherman, led an expedition in which he sailed a traditional Polynesian boat across 2,500 miles of ocean from Hawaii to Tahiti. The Polynesian Voyaging Society had organised the expedition. Its purpose was to find out if seafarers in the distant past could have found their way from one island to the other without navigational instruments, or whether the islands had been populated by accident. At the time, Mau was the only man alive who knew how to navigate just by observing the stars, the wind and the sea. He had never before sailed to Tahiti, which was a long way to the south. However, he understood how the wind and the sea behave around islands, so he was confident he could find his way. The voyage took him and his crew a month to complete and he did it without a compass or charts.

His grandfather began the task of teaching him how to navigate when he was still a baby. He showed him pools of water on the beach to teach him how the behaviour of the waves and wind changed in different places. Later, Mau used a circle of stones to memorise the positions of the stars. Each stone was laid out in the sand to represent a star. The voyage proved that Hawaii's first inhabitants came in small boats and navigated by reading the sea and the stars. Mau himself became a keen teacher, passing on his traditional secrets to people of other cultures so that his knowledge would not be lost. He explained the positions of the stars to his students, but he allowed them to write things down because he knew they would never be able to remember everything as he had done.

True/False/Not Given

Do the following statements agree with the information given in Reading Passage ?

In boxes 1-7 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1 At the time of his voyage, Mau had unique navigational skills.

2 Mau was familiar with the sea around Tahiti.

3 Mau thought it would be difficult to use a compass and charts.

4 Mau's grandfather was his only teacher.

5 Mau used stones to learn where each star was situated in the sky.

6 The first inhabitants of Hawaii could read and write.

7 Mau expected his students to memorise the positions of the stars.

Australian culture and culture shock

Sometimes work, study or a sense of adventure take us out of our familiar surroundings to go and live in a different culture. The experience can be difficult, even shocking.

Almost everyone who studies, lives or works abroad has problems adjusting to a new culture. This response is commonly referred to as 'culture shock'. Culture shock can be defined as 'the physical and emotional discomfort a person experiences when entering a culture different from their own'(Weaver,1993).

For people moving to Australia, Price(2001)has identified certain values which may give rise to culture shock. Firstly, he argues that Australians place a high value on independence and personal choice. This means that a teacher or course tutor will not tell students what to do, but will give them a number of options and suggest they work out which one is the best in their circumstances. It also means that they are expected to take action if something goes wrong and seek out resources and support for themselves.

Australians are also prepared to accept a range of opinions rather than believing there is one truth. This means that in an educational setting, students will be expected to form their own opinions and defend the reasons for that point of view and the evidence for it.

Price also comments that Australians are uncomfortable with differences in status and hence idealise the idea of treating everyone equally. An illustration of this is that most adult Australians call each other by their first names. This concern with equality means that Australians are uncomfortable taking anything too seriously and are even ready to joke about themselves.

Australians believe that life should have a balance between work and leisure time. As a consequence, some students may be critical of others who they perceive as doing nothing but study. Australian notions of privacy mean that areas such as financial matters, appearance and relationships are only discussed with close friends. While people may volunteer such information, they may resent someone actually asking them unless the friendship is firmly established. Even then, it is considered very impolite to ask someone what they earn. With older people, it is also rude to ask how old they are, why they are not married or why they do not have children. It is also impolite to ask people how much they have paid for something, unless there is a very good reason for asking.

Kohls(1996)describes culture shock as a process of change marked by four basic stages. During the first stage, the new arrival is excited to be in a new place, so this is often referred to as the "honeymoon" stage. Like a tourist, they are intrigued by all the new sights and sounds, new smells and tastes of their surroundings. They may have some problems, but usually they accept them as just part of the novelty. At this point, it is the similarities that stand out, and it seems to the newcomer that people everywhere and their way of life are very much alike. This period of euphoria may last from a couple of weeks to a month, but the letdown is inevitable.

During the second stage, known as the 'rejection' stage, the newcomer starts to experience difficulties due to the differences between the new culture and the way they were accustomed to living. The initial enthusiasm turns into irritation, frustration, anger and depression, and these feelings may have the effect of people rejecting the new culture so that they notice only the things that cause them trouble, which they then complain about. In addition, they may feel homesick, bored, withdrawn and irritable during this period as well.

Fortunately, most people gradually learn to adapt to the new culture and move on to the third stage, known as 'adjustment and reorientation'. During this stage a transition occurs to a new optimistic attitude. As the newcomer begins to understand more of the new culture, they are able to interpret some of the subtle cultural clues which passed by unnoticed earlier. Now things make more sense and the culture seems more familiar. As a result, they begin to develop problem-solving skills, and feelings of disorientation and anxiety no longer affect them.

In Kohls's model, in the fourth stage, newcomers undergo a process of adaptation. They have settled into the new culture, and this results in a feeling of direction and self-confidence. They have accepted the new food, drinks, habits and customs and may even find themselves enjoying some of the very customs that bothered them so much previously. In addition, they realise that the new culture has good and bad things to offer and that no way is really better than another, just different.

True/False/Not Given

Do the following statements agree with the information given in Reading Passage ?

In boxes 1-6 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1 Australian teachers will suggest alternatives to students rather than offer one solution.

2 In Australia, teachers will show interest in students' personal circumstances.

3 Australians use people's first names so that everyone feels their status is similar.

4 Students who study all the time may receive positive comments from their colleagues.

5 It is acceptable to discuss financial issues with people you do not know well.

6 Younger Australians tend to be friendlier than older Australians.

Complete the table below.

Choose ***NO MORE THAN TWO WORDS*** *from the passage for each answer.*

	name	**newcomers' reaction to problems**
Stage1		They notice the ____________________ between different nationalities and cultures. They may experience this stage for up to ______________ .
Stage2	Rejection	They reject the new culture and lose the _________they had at the beginning.

Stage3		They can understand some ________ which they had not previously observed. They learn ____________ for dealing with difficulties.
Stage4		They enjoy some of the customs that annoyed them before.

Lesson Three

Objectives

- ◇ labelling
- ◇ Yes/ No / Not given

Passage1

The way the brain buys

Supermarkets take great care over the way the goods they sett are arranged. This is because they know a lot about bow to persuade people to buy thing.

When you enter a supermarket, it takes some time for the mind to get into a shopping mode. This is why the area immediately inside the entrance of a supermarket is known as the 'decompression zone'. People need to slow down and take stock of the surroundings, even if they are returning customer. Supermarkets do not expect to sell much here, so it tends to be used more for promotion. So the large items piled up here are designed to suggest that there are bargains further inside the store, and shoppers are not necessarily expected to buy them. Walmart, the world's biggest retailer, famously employs ‘greeters’ at the entrance to its stores. A friendly welcome is said to cut shoplifting. It is harder to steal from nice people.

Immediately to the left in many supermarkets is a ‘chill zone’, where customers can enjoy browsing magazines, books and DVDs. This is intended to tempt unplanned purchases and slow customers down. But people who just want to do their shopping quickly will keep walking ahead, and the first thing they come to is the fresh fruit and vegetables section. However, for shoppers, this makes no sense. Fruit and vegetables can be easily damaged, so they should be bought at the end, not the beginning, of a shopping trip. But psychology is at work here selecting these items makes people feel good, so they feel less witty about reaching for less healthy food later on.
Shoppers already know that everyday items, like milk, are invariably placed towards the back of a store to provide more opportunity to tempt customers to buy things which are not on their shopping list. This is why pharmacies are also generally at the back. But supermarkets know shoppers know this, so they use other tricks, like placing popular items halfway along a section so that people have to walk all along the aisle looking for them. The idea is to boost 'dwell time’: the length of time people spend in a store.

Having walked to the end of the fruit-and-vegetable aisle, shoppers arrive at counters of prepared food, the fishmonger, the butcher and the deli. Then there is the in-store bakery, which can be smelt before it is seen. Even small supermarkets now use in-store bakeries. Mostly these bake pre-prepared items and frozen ingredients which have been delivered to the supermarket previously, and their numbers have increased, even though central bakeries that deliver to a number of stores are much more efficient. They do it for the smell of freshly baked bread, which arouses people's appetites and thus encourages them to purchase not just bread but also other food, including ready meals.

Retailers and producers talk a lot about the 'moment of truth'. This is not a philosophical idea, but the point when people standing in the aisle decide to buy something and reach to get it. At the instant coffee section, for example, branded products from the big producers are arranged at eye level while cheaper ones are lower down, along with the supermarket's own-label products.

But shelf positioning is fiercely fought over, not just by those trying to sell goods, but also by those arguing over how best to manipulate shoppers. While many stores reckon eye level is the top spot, some think a little higher is better. Others think goods displayed at the end of aisles sell the most because they have the greatest visibility. To be on the right-hand side of an eye-level selection is often considered the very best place, because most people are right-handed and most people's eyes drift rightwards. Some supermarkets reserve that for their most expensive own-label goods.

Scott Bearse, a retail expert with Deloitte Consulting in Boston, Massachusetts, has led projects observing and questioning tens of thousands of customers about how they feel about shopping. People say they leave shops empty-handed more often because they are 'unable to decide' than because prices are too high, says Mr Bearse. Getting customers to try something is one of the best ways of getting them to buy, adds Mr Bearse. Deloitte found that customers who use fitting rooms in order to try on clothes buy the product they are considering at a rate of 85% compared with 58%for those that do not do so.

Often a customer struggling to decide which of two items is best ends up not buying either. In order to avoid a situation where a customer decides not to buy either product, a third 'decoy' item, which is not quite as good as the other two, is placed beside them to make the choice easier and more pleasurable. Happier customers are more likely to buy.

True/False/Not Given

Do the following statements agree with the information given in Reading Passage ?

In boxes 5-10 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

5. The 'greeters' at Walmart increase sales.
6. People feel better about their shopping if they buy fruit and vegetables before they buy other food.
7. In-store bakeries produce a wider range of products than central bakeries.

8. Supermarkets find right-handed people easier to persuade than left-handed people.
9. The most frequent reason for leaving shops without buying something is price.
10. 'Decoy' items are products which the store expects customers to choose.

Complete the flow chart below.

Choose ***NO MORE THAN TWO WORDS*** *from the passage for each answer.*

In-store bread production process

prepared

The supermarket is sent **11**______________________and other items which have been

Baking bread in-store produces an aroma.

Shoppers' **12**______________________________are stimulated.

earlier.

They are then keener to buy food, including bread and

13________ .

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26, which are based on Reading Passage 2 below.

Nature or Nurture?

A A few years ago, in one of the most fasc1nat1ng and disturbing experiments in behavioural psychology, Stanley Milgram of Yale University tested 40 subjects from all walks of life for their willingness to obey instructions given by a 'leader' in a situation in which the subjects might feel a personal distaste for the actions they were called upon to perform. Specifically, Milgram told each volunteer 'teacher-subject' that the experiment was in the noble cause of education, and was designed to test whether or not punishing pupils for their mistakes would have a positive effect on the pupils' ability to learn.

B Milgram's experimental set-up involved plac1ng the teacher-subject before a panel of thirty switches with labels ranging from '15 volts of electricity (slight shock)' to '450 volts (danger- severe shock)' in steps of 15 volts each. The teacher-subject was told that whenever the pupil gave the wrong answer to a question, a shock was to be administered, beginning at the lowest level and increasing in severity with each successive wrong answer. The supposed 'pupil' was in reality an actor hired by Milgram to simulate receiving the shocks by emitting a spectrum of groans, screams and writhings together with an assortment of statements and expletives denounc1ng both the experiment and the experimenter. Milgram told the teacher-subject to 1gnore the reactions of the pupil, and to administer whatever level of shock was called for, as per the rule governing the experimental situation of the moment.

C As the experiment unfolded, the pupil would deliberately give the wrong answers to questions posed by the teacher, thereby bringing on various electrical punishments, even up to the danger level of 300 volts and beyond. Many of the teacher-subjects balked at administering the higher levels of punishment, and turned to Milgram with questioning looks and/or complaints about continuing the experiment. In these situations, Milgram calmly explained that the teacher-subject was to ignore the pup11's cries tor mercy and carry on with the experiment. If the subject was still reluctant to proceed, Milgram said that it was important for the sake of the experiment that the procedure be followed through to the end. His final argument was, 'You have no other choice. You must go on.' What Milgram was trying to discover was the number of teacher-subjects who would be willing to administer the highest levels of shock, even in the face of strong personal and moral revulsion against the rules and conditions of the experiment.

D. Prior to carrying out the experiment, Milgram explained his idea to a group of psychiatrists and asked them to predict the average percentage of people in an ordinary population who would be willing to administer the highest shock level of 450 volts. The overwhelming consensus was that virtually all the teacher-subjects would refuse to obey the experimenter. The psychiatrists felt that 'most subjects would not go beyond 150 volts' and they further anticipated that only tour per cent would go up to 300 volts. Furthermore. they thought that only a lunatic fringe of about one in 1 .000 would give the highest shock of 450 volts.

E What were the actual results? Well over 60 per cent of the teacher-subjects continued to obey Milgram up to the 450-volt limit! In repetitions of the experiment in other countries, the percentage of obedient teacher-subjects was even higher. reaching 85 per cent in one country. How can we possibly account for this vast discrepancy between what calm, rational. knowledgeable people predict in the comfort of their study and what pressured, flustered, but cooperative 'teachers' actually do 1n the laboratory of real life?

F One's first inclination might be to argue that there must be some sort of built-in animal aggression instinct that was activated by the experiment, and that Milgram's teacher□subjects were just following a genetic need to discharge this pent-up primal urge onto the pupil by administering the electrical shock. A modern hard-core sociobiologist might even go so far as to claim that this aggressive instinct evolved as an advantageous trait. having been of surv1val value to our ancestors in their struggle against the hardships of life on the plains and in the caves, ultimately finding its way into our genetic make-up as a remnant of our ancient animal ways.

G An alternative to this notion of genetic programming is to see the teacher-subjects' actions as a result of the social environment under which the experiment was carried out. As Milgram himself pointed out. 'Most subjects 1n the expenment see their behaviour in a larger context that is benevolent and useful to society- the pursuit of scientific truth. The psychological laboratory has a strong claim to legitimacy and evokes trust and confidence in those who perform there. An action such as shocking a victim, which in isolation appears evil, acquires a completely different meaning when placed in this setting.'

H Thus. in this explanation the subject merges his unique personality and personal and moral code with that of larger institutional structures, surrendering individual properties like loyalty, self-sacrifice and discipline to the service of malevolent systems of authority.

I Here we have two radically different explanations for why so many teacher-subjects were willing to forgo their sense of personal responsibility for the sake of an institutional authority figure. The problem for biologists, psychologists and anthropologists is to sort out which of these two polar explanations is more plausible. This, in essence. is the problem of modern sociobiology -to discover the degree to which hard-wired genetic programming dictates, or at least strongly b1ases, the interaction of animals and humans with their environment, that is. their behaviour. Put another way, sociobiology is concerned with elucidating the biological basis of all behaviour.

Questions 14-19

Reading Passage 2 has nine paragraphs, A-1.

Which paragraph contains the following information?

Write the correct letter A-1 in boxes 14-19 on your answer sheet.

14 a biological explanation of the teacher-subjects' behaviour

15 the explanation Milgram gave the teacher-subjects for the experiment

16 the identity of the pupils

17 the expected statistical outcome

18 the general aim of sociobiological study

19 the way Milgram persuaded the teacher-subjects to continue

Questions 20-22

Choose the correct letter. A, B. Cor D.

Write your answers in boxes 20-22 on your answer sheet.

20 The teacher-subjects were told that they were testing whether

A a 450-volt shock was dangerous.

B punishment helps learning.

C the pupils were honest.

D they were suited to teaching.

21 The teacher-subjects were instructed to

A stop when a pupil asked them to.

B denounce pupils who made mistakes.

C reduce the shock level after a correct answer.

D give punishment according to a rule.

22 Before the experiment took place the psychiatrists

A believed that a shock of 150 volts was too dangerous.

B failed to agree on how the teacher-subjects would respond to instructions.

C underestimated the teacher-subjects' willingness to comply with experimental procedure.

D thought that many of the teacher-subjects would.administer a shock of 450 volts.

AUSTRALIA'S SPORTING SUCCESS

A They play hard, they play often, and they play to win. Australian sports teams win more than their fair share of titles, demolishing rivals with seeming ease. How do they do it? A big part of the secret is an extensive and expensive network of sporting academies underpinned by science and medicine . At the Australian Institute of Sport (AIS), hundreds of youngsters and pros live and train under the eyes of coaches. Another body, the Australian Sports Commission (ASC), finances programs of excellence in a total of 96 sports for thousands of sportsmen and women. Both provide intensive coaching, training facilities and nutritional advice.

B Inside the academies, science takes centre stage. The AIS employs more than I 00 sports scientists and doctors, and collaborates with scores of others in universities and research centres. AIS scientists work across a number of sports, applying skills learned in one - such as building muscle strength in golfers- to others, such as swimming and squash. They are backed up by technicians who design instruments to collect data from athletes. They all focus on one aim: winning. 'We can't waste our time looking at ethereal scientific questions that don't help the coach work with an athlete and improve performance,' says Peter Fricker; chief of science at AIS. ·

C A lot of their work comes down to measurement - everything from the exact angle of a swimmer's dive to the second-by-second power output of a cyclist. This data is used to wring improvements out of athletes. The focus is on individuals, tweaking performances to squeeze an extra hundredth of a second here , an extra millimetre there. No gain is too slight to bother with. it's the tiny, gradual improvements that add up to world-beating results. To demonstrate how the system works, Bruce Mason at.tyS shows off the prototype of a 3D analysis tool for studying swimmers. A wire-frame model of a champion swimmer slices through the water; her arms moving in slow motion. Looking side-on, Mason measures the distance between strokes. From above, he analyses how her spine swivels. When fully developed, this system will enable him to build a biomechamical profile for coaches to use to help budding swimmers. Mason's contribution to sport also includes the development of the SWAN (SWimming ANalysis) system now used in Australian national competitions. it collects images from digital cameras running at 50 frames a second and breaks down each part of a swimmer's performance into factors that can be analysed individually - stroke length, stroke frequency, average duration of each stroke, velocity, start, lap and finish times, and so on. At the end of each race, SWAN spits out data on each swimmer.

D Take a look,' says Mason, pulling out a sheet of data. He points out the data on the swimmers in second and third place, which shows that the one who finished third actually swam faster. So why did he finish 35 hundredths of a second down? 'His turn times were 44 hundredths of a second behind the other guy.' says Mason. 'If he can improve on his turns, he can do much better.' This is the kind of accuracy that AIS scientists' research is bringing to a range of sports. With the Cooperative Research Centre for Micro Technology in Melbourne, they are developing unobtrusive sensors that will be embedded in an athlete's clothes or running shoes to monitor heart rate, sweating, heat production or any other factor that might have an impact on an athlete's ability to run. There's more to it than simply measuring performance. Fricker gives the example of athletes who may be down with coughs and colds I I or 12 times a year. After years of experimentation, AIS and the University of Newcastle in New South Wales developed a test that measures how much of the immune-system protein immunoglobulin A is present in athletes' saliva. If lgA levels suddenly fall below a certain level, training is eased or dropped altogether. Soon, lgA levels start rising again, and the danger passes. Since the tests were introduced, AIS athletes in all sports have been remarkably successful at staying healthy.

E Using data is a complex business. Well before a championship, sports scientists and coaches start to prepare the athlete by developing a 'competition model', based on what they expect will be the winning times. 'You design the model to make *that* time,' says Mason. 'A start of *this* much, each free-swimming period has to be *this* fast, with a certain stroke frequency and stroke length, with turns done in *these* times.' All the training is then geared towards making the athlete hit those targets, both overall and for each segment of the race. Techniques like these have transformed Australia into arguably the world's most successful sporting nation.

F Of course, there's nothing to stop other countries copying - and many have tried. Some years ago, the AIS unveiled coolant-lined jackets for endurance athletes. At the Atlanta Olympic Games in 1996, these sliced as much as two per cent off cyclists' and rowers' times. Now everyone uses them. The same has happened to the 'altitude tent', developed by AIS to replicate the effect of altitude training at sea level. But Australia's success story is about more than easily copied technological fixes, and up to now no nation has replicated its all-encompassing system.

Questions 1-7

Reading Passage 1 has six paragraphs, A-F.

Which paragraph contains the following information?

Write the correct letter, A-F, in boxes 1-7 on your answer sheet.

NB You may use any letter more than once.

1 a reference to the exchange of expertise between different sports

2 an explanation of how visual imaging is employed in investigations

3 a reason for narrowing the scope of research activity

4 how some AIS ideas have been reproduced

5 how obstacles to optimum achievement can be investigated

6 an overview of the funded support of athletes

7 how performance requirements are calculated before an event

Questions 8-11

Classify the following techniques according to whether the writer states they

A *are currently exclusively used by Australians*

B *will be used in the future by Australians*

C *are currently used by both Australians and their rivals*

Write the correct letter, A, B or C, in boxes 8-11 on your answer sheet.

8 cameras

9 sensors

10 protein tests

11 altitude tents

Questions 12 and 13

Answer the questions below. Choose ***NO MORE THAN THREE WORDS AND/OR A NUMBER*** *from the passage for each answer.*

Write your answers in boxes 12 and 13 on your answer sheet.

12 What is produced to help an athlete plan their performance in an event?

13 By how much did some cyclists' performance improve at the 1996 Olympic Games?

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40, which are based on Reading Passage below.

Numeration

One of the first great intellectual feats of a young child is learning how to talk, closely followed by learning how to count. From earliest childhood we are so bound up with our system of numeration that it is a feat of imagination to consider the problems faced by early humans who had not yet developed this facility. Careful consideration of our system of numeration leads to the conviction that, rather than being a facility that comes naturally to a person, it is one of the great and remarkable achievements of the human race.

It is impossible to learn the sequence of events that led to our developing the concept of number. Even the earliest of tribes had a system of numeration that, if not advanced, was sufficient for the tasks that they had to perform. Our ancestors had little use for actual numbers; instead their considerations would have been more of the kind /s *this enough?* rather than *How many?* when they were engaged in food gathering, for example. However, when early humans first began to reflect on the nature of things around them, they discovered that they needed an idea of number simply to keep their thoughts in order. As they began to settle, grow plants and herd animals, the need for a sophisticated number system became paramount. It will never be known how and when this numeration ability developed, but it is certain that numeration was well developed by the time humans had formed even semi□permanent settlements.

Evidence of early stages of arithmetic and numeration can be readily found . The indigenous peoples of Tasmania were only able to count one, two, many; those of South Africa counted one, two, two and one, two twos, two twos and one, and so on. But in real situations the number and words are often accompanied by gestures to help resolve any confusion . For example, when using the one, two, many type of system, the word many would mean, Look at my hands and see how many fingers I am showing you. This basic approach is limited in the range of numbers that it can express, but this range will generally suffice when dealing with the simpler aspects of human existence.

The lack of ability of some cultures to deal with large numbers is not really surprising. European languages, when traced back to their earlier version, are very poor in number words and expressions. The ancient Gothic word for ten, tachund, is used to express the number 1 00 as tachund tachund. By the seventh century, the word teon had become interchangeable with the tdchund or hund of the Anglo-Saxon language, and so 100 was denoted as hund teontig, or ten times ten . The average person in the seventh century in Europe was not as familiar with numbers as we are today. In fact, to qualify as a witness in a court of law a man had to be able to count to nine!

Perhaps the most fundamental step in developing a sense of number is not the ability to count, but rather to see that a number is really an abstract idea instead of a simple attachment to a group of particular objects. it must have been within the grasp of the earliest humans to conceive that four birds are distinct from two birds; however, it is not an elementary step to associate the number, as connected with four birds, to the number , as connected with four rocks. Associating a number as one of the qualities of a specific object is a great hindrance to the development of a true number sense. When the number can be registered in the mind as a specific word, independent of the object being referenced, the individual is ready to take the first step toward the development of a national system for numbers and, from there, to

arithmetic.

Traces of the very first stages in the development of numeration can be seen in several living languages today. The numeration system of the Tsimshian language in British Columbia contains seven distinct sets of words for numbers according to the class of the item being counted : for counting flat objects and animals, for round objects and time, for people, for long objects and trees, for canoes, for measures, and for counting when no particular object is being numerated . It seems that the last is a later development while the first six groups show the relics of an older system. This diversity of number names can also be found in some widely used languages such as Japanese.

Intermixed with the development of a number sense is the development of an ability to count. Counting is not directly related to the formation of a number concept because it is possible to count by matching the items being counted against a group of pebbles, grains of corn, or the counter's fingers . These aids would have been indispensable to very early people who would have found the process impossible without some form of mechanical aid. Such aids, while different, are still used even by the most educated in today's society due to their convenience. All counting ultimately involves reference to something other than the things being counted. At first it may have been grains or pebbles but now it is a memorised sequence of words that happen to be the names of the numbers.

Questions 27-31

Complete each sentence with the correct ending, A-G, below. Write the correct letter, A-G, in boxes 27-31 on your answer sheet.

27 A developed system of numbering

28 An additional hand signal

29 In seventh-century Europe, the ability to count to a certain number

30 Thinking about numbers as concepts separate from physical objects

31 Expressing number differently according to class of item

A was necessary in order to fulfil a civic role.

B was necessary when people began farming.

C was necessary for the development of arithmetic.

D persists in all societies.

E was used when the range of number words was restricted.

F can be traced back to early European languages.

G was a characteristic of early numeration systems.

Traffic jams-no end in sight

There are no easy answers to the problems of traffic congestion.

A. Traffic congestion affects people throughout the world Traffic jams cause smog in dozens of cities across both the developed and developing world. In the US, commuters spend an average of a full working week each year sitting in traffic jams, according to the Texas Transportation Institute.While alternative ways of getting around are available, most people still choose their cars because they are looking for convenience, comfort and privacy.

B. The most promising technique for reducing city traffic is called congestion pricing, whereby cities charge a toll to enter certain parts of town at certain times of day. In theory, if the toll is high enough, some drivers will cancel their trips or go by bus or train. And in practice it seems to work: Singapore, London and Stockholm have reduced traffic and pollution in city centres thanks to congestion pricing.

C. Another way to reduce rush-hour traffic is for employers to implement flexitime, which lets employees travel to and from work at off-peak traffic times to avoid the rush hour. Those who have to travel during busy times can do their part by sharing cars. Employers can also allow more staff to telecommute [work from home] so as to keep more cars off the road altogether.

D. Some urban planners still believe that the best way to ease traffic congestion is to build more roads, especially roads that can take drivers around or over crowded city streets.But such techniques do not really keep cars off the road; they only accommodate more of them.

E. Other, more forward-thinking, planners know that more and more drivers and cars are taking to the roads every day, and they are unwilling to encourage more private automobiles when public transport is so much better both for people and the environment. For this reason, the American government has decided to spend some $7billion on helping to increase capacity on public-transport systems and upgrade them with more efficient technologies. But environmentalists complain that such funding is tiny compared to the$50 billion being spent on roads and bridges.

Matching headings

*The reading passage has six paragraphs, **A-E**. Choose the correct heading for each paragraph from the list of headings below.*

List of Headings	
i	A solution which is no solution
ii	Changing working practices
iii	Closing city centres to traffic
iv	Making cars more environmentally friendly
v	Not doing enough
vi	Paying to get in
vii	A global problem

1 Paragraph A................

2 Paragraph B................

3 Paragraph C................

4 Paragraph D................

5 Paragraph E................